Lecture 13

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Lecture 13

Engineering Mathematics III

ENGE702

Lecture 13: Systems of Differential Equations

Systems of Differential Equations

In this lecture we will look at **systems of linear differential equations** with constant coefficients. In this course we will only consider systems with two equations. They will be of the form:

$$y_1' = a_{11}y_1 + a_{12}y_2$$
$$y_2' = a_{21}y_1 + a_{22}y_2$$

Note that this can be written in the form:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

which we often abbreviate as

$$\mathbf{y}' = A\mathbf{y}$$

To solve two dependant variable y_1 and y_2 in a system of linear equations.

$$\begin{cases} y_1' = a_{11}y_1 + a_{12}y_2 \\ y_2' = a_{21}y_1 + a_{22}y_2 \end{cases}$$

$$\Rightarrow \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

in matrix shorthand notation.

$$y' = Ay$$

Systems of Differential Equations

Solving this system means finding functions y_1 and y_2 that satisfy the system of equations.

We have a system of differential equations

$$\mathbf{y}' = A\mathbf{y}.$$

We are dealing with vectors and matrices now, but this looks similar to the differential equation

$$y' = ky$$

for which the solution was

$$y = Ce^{kt}.$$

We will assume it is something like this, and see if we can make it work.

Aim is to solve the system of ODEs $y' = Ay$ to obtain the unknown functions y_1 and y_2.

If there is only one unknown function y to solve, we know the general solution has the form $y(t) = C \cdot e^{kt}$.

similar to the scalar case.

Systems of Differential Equations

We will assume a solution to the system is of the form

$$\boxed{\mathbf{y} = \mathbf{x}e^{\lambda t}}$$

We will assume a solution to the system is of the form

$$\boxed{\mathbf{y} = \mathbf{x}e^{\lambda t}}$$

for some vector $\mathbf{x}$ and some constant λ. Then

differentiate from the assumed solution.

$$\mathbf{y}' = \lambda \mathbf{x}e^{\lambda t},$$

but we also know from the formulation of our problem that $\mathbf{y}' = A\mathbf{y}$, so we put that together with $\mathbf{y} = \mathbf{x}e^{\lambda t}$, and we have

Substitute the form of y and y' into the system of ODEs

$$\mathbf{y}' = A\mathbf{x}e^{\lambda t}$$

$$\Downarrow$$

$$\lambda x e^{\lambda t} = A x e^{\lambda t}$$

So

$$A\mathbf{x}e^{\lambda t} = \lambda \mathbf{x}e^{\lambda t}$$

so we need a constant λ and a vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda \mathbf{x}.$$

$\Rightarrow Ax = \lambda x$, so x are the eigenvectors of A and λ are the eigenvalues of A.

Systems of Differential Equations

This is exactly finding an eigenvalue and eigenvector of A! Note for two equations our general solution will have two solutions, these come from the two eigenvalues and associated eigenvectors.

Our general solution to our system of equations is then

$$\mathbf{y} = c_1\mathbf{x}_1 e^{\lambda_1 t} + c_2\mathbf{x}_2 e^{\lambda_2 t}$$

where $\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ x_{21} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_{12} \\ x_{22} \end{pmatrix}$ are eigenvectors of matrix A corresponding to eigenvalues λ_1 and λ_2 respectively, and c_1 and c_2 are arbitrary constants (which we could find if we had initial conditions).

This vector equation can then read:

$$y_1 = c_1 x_{11} e^{\lambda_1 t} + c_2 x_{12} e^{\lambda_2 t}$$
$$y_2 = c_1 x_{21} e^{\lambda_1 t} + c_2 x_{22} e^{\lambda_2 t}$$

Example: Solve the system of ODEs for y_1 and y_2.

$$y_1' = 7y_1 - y_2$$
$$y_2' = 3y_1 + 3y_2$$

General steps: ① write the equations in matrix form $y' = Ay$.

② Find the eigenvalues and eigenvectors of the matrix A: λ_1, λ_2, x_1, x_2

③ Write down the general solution as $y(t) = c_1 x_1 e^{\lambda_1 t} + c_2 x_2 e^{\lambda_2 t}$, $c_1, c_2 \in \mathbb{R}$

④ Apply initial conditions (if given) to determine the arbitrary constants c_1, c_2

Systems of Differential Equations

Example: Find the general solution to the following system of differential equations:

$$y_1' = 7y_1 - y_2$$
$$y_2' = 3y_1 + 3y_2$$

Step ① rewrite in matrix form:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

so $A = \begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}$.

step ② Find eigenvalues and eigenvectors of A.

i) Solve $\det(A - \lambda I) = 0$ for λ.

$$\det(A - \lambda I) = \det\left(\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right)$$

$$= \det\begin{pmatrix} 7-\lambda & -1 \\ 3 & 3-\lambda \end{pmatrix} = (7-\lambda)(3-\lambda) - 3\times(-1) = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 21 + 3 = 0 \Rightarrow \lambda^2 - 10\lambda + 24 = 0$$

Use the quadratic formula:

$$\lambda = \frac{-(-10) \pm \sqrt{(-10)^2 - 4\times1\times24}}{2} = \frac{10 \pm \sqrt{4}}{2}$$

$$\Rightarrow \lambda_1 = 6 \text{ and } \lambda_2 = 4.$$

ii) Solve $(A - \lambda I)x = 0$ for the eigenvectors x.

For $\lambda_1 = 6$: $(A - 6I)x = \left[\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} - \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}\right]x = \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2$$

The eigenvector for $\lambda_1 = 6$ is $v_1 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_2 \end{pmatrix} = x_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $x_2 \in \mathbb{R}$

For $\lambda_2 = 4$: $(A - 4I)x = \left(\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\right)x = \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Systems of Differential Equations

Example: Find the general solution to the following system of differential equations:

$$y_1' = 7y_1 - y_2$$
$$y_2' = 3y_1 + 3y_2$$

We can express this as

$$\mathbf{y}' = \begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}\mathbf{y}.$$

Now we need to find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}$. Start by solving $\det(A - \lambda I) = 0$. We have

$$(7-\lambda)(3-\lambda) + 3 = 0$$
$$\lambda^2 - 10\lambda + 24 = 0$$

$$
\begin{aligned}
(7-\lambda)(3-\lambda)+3 &= 0 \\
\lambda^2 - 10\lambda + 24 &= 0
\end{aligned}
$$

$$\lambda = 6 \quad \text{or} \quad \lambda = 4$$

The eigenvector for $\lambda_1 = 6$ is $v_1 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}, x_2 \in \mathbb{R}$

For $\lambda_2 = 4$ $(A - 4I)x = \left(\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \right) x = \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Rightarrow 3x_1 - x_2 = 0 \Rightarrow x_2 = 3x_1$.

The eigenvector for $\lambda_2 = 4$ is $v_2 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ 3x_1 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix}, x_1 \in \mathbb{R}$.

Systems of Differential Equations

Now we can find the eigenvectors:

For $\lambda = 6$:

$$\begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 1$ will work. So $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is our first eigenvector.

For $\lambda = 4$:

$$\begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 3$ will work. So $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is our second eigenvector (for $\lambda_2 = 4$).

Step 3: The general solution of the ODE system is

$y = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = c_1 \cdot v_1 \cdot e^{\lambda_1 t} + c_2 v_2 \cdot e^{\lambda_2 t}$.

In our example,

$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = c_1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$.

$\Rightarrow y_1(t) = c_1 e^{6t} + c_2 e^{4t}$

$y_2(t) = c_1 e^{6t} + 3c_2 e^{4t}$.

Systems of Differential Equations

Our general solution will then be:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$$

Systems of Differential Equations

Example: Solve the initial value problem:

$$
\begin{aligned}
y_1' &= 7y_1 - y_2 \\
y_2' &= 3y_1 + 3y_2
\end{aligned}
$$

where $y_1(0) = 1$, $y_2(0) = 0$.

Step 4: Use the ICs to determine the arbitrary constants c_1 and c_2

$y = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$.

At $t=0$ $y(0) = c_1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot e^0 + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\Rightarrow c_1 + c_2 = 1$ $\quad -3c_2 + c_2 = 1 \Rightarrow c_2 = -\frac{1}{2}$

$c_1 + 3c_2 = 0 \Rightarrow c_1 = -3c_2$ $\quad c_1 = \frac{3}{2}$

So the particular solution is $y = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} - \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$.

Systems of Differential Equations

Example: Solve the initial value problem:

$$y_1' = 7y_1 - y_2$$
$$y_2' = 3y_1 + 3y_2$$

where $y_1(0) = 1$, $y_2(0) = 0$.

As usual, to solve an initial value problem we would need to first find the general solution, and then apply the initial conditions to get values for the constants.

In this case the system is the same as the last one, so we can just use the general solution we just found:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

We know that $y_1(0) = 1$ and $y_2(0) = 0$, so applying this we get:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{(6)(0)} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{(4)(0)}.$$

Systems of Differential Equations

This gives us

$$1 = c_1 + c_2$$
$$0 = c_1 + 3c_2$$

Solving this system of equations gives $c_1 = \frac{3}{2}$ and $c_2 = -\frac{1}{2}$, so our particular solution becomes:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} - \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

Systems of Differential Equations

Exercise: Solve the following initial value problem:

$$y_1' = y_1 - y_2$$
$$y_2' = 2y_1 - 2y_2,$$

where $y_1(0) = 1$, $y_2(0) = 0$.

Text Book References

Chapters 4.0, 4.1